Sahaj Bhadja

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Summary

Software Engineer with 2+ years writing and debugging **Java** and **Python** services that issue and serve HTTP requests, building **payment processing** integrations with refunds and chargebacks, REST APIs, data models, and cloud deployments in Agile teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed relational **data models** with row-level security and indexing for a 500+ user platform on Azure SQL and PostgreSQL, tuning queries to cut dashboard latency from 1.2s to 350ms.
- Built Python integration pipelines with reconciliation checks and **data warehouse**-style SQL transformations across PostgreSQL and SQL Server, sustaining 99%+ reliability over 5K+ monthly jobs through CI/CD-backed production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built Python Flask services that **issue and serve HTTP requests** to external model providers with timeout handling and automatic fallbacks, serving 150+ instructors and 10K+ daily requests on AWS.
- **Debugged** production timeouts and failures across third-party APIs, adding multithreaded batch processing and per-call usage logging that kept responses stable and cut instructor feedback turnaround 60% in production.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built object-oriented **Java** and Python modules for large-scale external API integrations, covered by 20+ unit and **integration tests** with Mockito and PyTest in nightly Jenkins pipelines across Windows, Linux, and macOS.
- Improved REST API performance with pagination, field filtering, and caching, cutting response times 30%, and redesigned AWS Glue and S3 ETL for 68% higher efficiency in Agile sprints using Jira and Git.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **Java Spring Boot** REST microservices for invoices, payments, and billing cycles with payment gateway integration, syncing 2M+ records across locations through real-time WebSocket updates for every site.
- Modeled Azure SQL tables for invoices, payments, and customer records, and shipped through Jenkins CI/CD with unit, integration, and end-to-end tests, reaching 95%+ successful **build validation** across release runs.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built PCI-compliant multi-currency **card payment** infrastructure across cards, UPI, wallets, and PayPal, with fraud detection and webhooks handling payment status, refunds, disputes, and **chargebacks** across 50K+ booking records.
- Kept **payment status** consistent with idempotent webhook processing and transactional locking, cutting manual refund handling about 70%, alongside hourly Spark revenue analytics across 50K+ booking and payment records.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL
Payments: Payment Processing, Card Payments, Refunds, Chargebacks, Disputes, Webhooks, Idempotency, PCI Compliance
APIs & Backend: REST APIs, HTTP, Spring Boot, Flask, FastAPI, Node.js, Microservices
Data & Cloud: PostgreSQL, SQL Server, Azure SQL, Data Modeling, ETL, Spark, AWS, Azure, Docker
Practices: Unit Testing, Integration Testing, Mockito, PyTest, Debugging, Agile, Jira, Git, Jenkins, CI/CD, Linux

Projects

ISO 8583 Card Authorization Simulator | *Java, ISO 8583, Netty, PostgreSQL, JUnit*

- Designed a **Java** service that builds and parses **ISO 8583** authorization and reversal messages, routes them to a simulated card network over TCP, and records responses for reconciliation.

Card Clearing Reconciliation Pipeline | *Python, SQL, BigQuery, Airflow*

- Designed a pipeline that matches **authorizations** to daily clearing files by transaction ID and amount, flags unmatched or partial captures, and loads results into **warehouse** tables for queries.

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, Kubernetes*

- Built a **fintech** platform with Plaid bank linking, a token-based firewall, and real-time transaction updates, supporting 50,000+ users and 1M+ transactions with Kubernetes CI/CD on the cloud.